

Data Collection and Preprocessing Phase

Date	16August 2026
Team ID	----
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing: -

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	• Schema Inspection: Examined dataset attributes including, Season, Crop, Area (Hectares), and Production. • Dataset Profiling: Assessed missing values, structural irregularities, and value distributions across 30+ states, 600+ districts, and 50+ crop varieties from 1997 to 2021.
Data Cleaning	• Null Value Handling: Imputed missing values in Production using regional median yield rates; dropped records missing essential spatial identifiers. • Outlier & Noise Reduction: Removed corrupt

	entries with negative area values or impossible yield spikes caused by data entry errors
Data Transformation	Calculated the standardized productivity metric $\text{Yield (Tones/Hectare)} = \frac{\text{Production}}{\text{Area}}.$ Normalization: Grouped low-frequency minor crops into broad categories (e.g., Pulses, Oilseeds, Cereals) for macro-level trend analysis.
Data Type Conversion	• Numerical Casts: Explicitly converted Area and Production to floating-point numeric types to allow statistical aggregations. • Temporal & Categorical Casting: Formatted Crop Year into standard integer date format and converted categorical text fields into strings for dynamic visualization filtering.
Column Splitting and Merging	Standardization: Cleaned and merged inconsistent state/district naming variations resulting from geographical boundary redefinitions over time. Attribute Parsing: Split combined seasonal text attributes (e.g., combining climate tags) into standardized season labels (Kharif, Rabi, Summer, Whole Year).

Data Modeling	❏ Star Schema Architecture: Created a central Fact Tab linked to look up Dimension Tables. ❏ Relationship Building: Established primary-foreign key relationships (1:N) between dimensions and facts to optimize query speed and drill-down performance in Tableau.
Save Processed Data	❏ Storage Format: Exported clean data to optimized .csv / .parquet formats and pushed structured tables to SQL database staging tables. ❏ Tableau Extraction: Created a hyper-extracted data source file (.hyper) ready for rapid visualization render in